Andy Do

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Education

Cornell University Ithaca, NY

Bachelor of Arts in Computer Science *May 2028*

Relevant Coursework: Machine Learning, Computer System Organization, Discrete Mathematics, Data structures and Algorithms, Linear Algebra, Databases, Functional Programming, Intro to Python

Experience

Fiserv Jun 2026 – Aug 2026

Software Engineer Intern *King of Prussia, PA*

- Enabled **20,000+** sales partners to get automatic status alerts as new businesses signed up for card payments by building a self-service notification tool with **Java** and **Spring**.
- Migrated a user payments app with **30+** downstream integrations from VMs to Kubernetes, authoring **11+** **Helm**-packaged Deployments, **HPAs**, and **VPAs** for automated **Argo CD GitOps** via **GitLab**
- Protected payment and signup data for a **50,000+** business network by generating secure signing keys and adding access controls across **8** services using **Spring Security**.

Lancaster Lebanon Intermediate Unit 13 Jun 2024 – Aug 2024

Software Engineer Intern *Lancaster, PA*

- Facilitated the delivery of special needs services to **3800+** preschoolers by building 2 full-stack web applications with **ASP.NET Core** (**C#**), **HTML**, **JavaScript** and **Bootstrap**.
- Managed data for **7000+** parents and caregivers by designing a **SQL** database and implementing backend **CRUD**/business logic for student, guardian, and referral records.
- Shipped scheduled software releases by collaborating with **8** teammates via **Git** in an **Agile development** environment using **Azure Repos** and **Azure Boards**.

Cornell Data Science November 2025 – Present

Machine Learning Engineer *Ithaca, NY*

- Built an autograd engine from scratch in **NumPy** with a team of 4, implementing **vectorized backpropagation** and **SGD**, and trained an **MLP** achieving **92%** accuracy on MNIST testing.
- Built image texture descriptors in **Python** with **OpenCV** and **scikit-image**, benchmarking **161** AI-edited images to identify regions where digital watermarks survive manipulation and presenting findings to **100+** members.

Projects

CodeLens | *Fiserv Hack Week Winner* June 2026

- Built a full stack **Chrome extension** that maps any live web app component to its backend code flow, Jira tickets, and Confluence docs on hover, helping **8** new interns learn their assigned codebases.
- Utilized a **LangChain** ReAct agent (**Claude Sonnet 4.5**) with a **ChromaDB** vector store for retrieval across **GitHub**, **Jira**, and **Confluence**, and **FastAPI** for the backend.

Loci | *Yale Hacks* March 2026

- Built a full-stack platform that turns raw text and PDFs into 3D memory palaces using **TanStack Start**, **React**, **TypeScript**, and **MongoDB Atlas**.
- Generated 3D concept objects by self-hosting the open-source **Microsoft TRELLIS** model on cloud GPUs and integrating **Hugging Face**, **Gradio**, and **R2/S3** storage.
- Created an interactive **Three.js** world viewer that places AI-generated objects into explorable rooms for spatial studying.

Technical Skills

Languages: Java, Python, C#, C, JavaScript, TypeScript, SQL, HTML, CSS, OCaml, Go
Frameworks: React, Node.js, NextJS, TanStack, JPA, Spring Boot, tRPC, Bootstrap, Flask, ASP.NET Core, SQLAlchemy, Prisma
Developer Tools: Git, Docker, VS Code, IntelliJ, AWS, Bruno, Postman, Kubernetes, ArgoCD, Helm, Agile development, HuggingFace, Cloudflare, Jira, Confluence
Libraries: Pandas, NumPy, Matplotlib, scikit-learn